Email Security Gateway Requirements

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links. The system should inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages. The platform should maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events. The solution should be secure, reliable, scalable, and manageable for university-wide deployment.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures and the SDLC plan are available in the supporting report.

2 Overall description

2.1 Product perspective

Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- Access permissions — FR-002
- inspect incoming and outgoing email traffic for phishing attempts, malicious attachments, spoofed messages, and suspicious links. — FR-001

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
User	students, faculty, staff	Uses permitted functions; each source class keeps its specified permissions.
Admin	administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- Thresholds for what constitutes a suspicious link or attachment are not defined
- Access controls for which administrators can approve messages are not defined
- Definition of scalability in terms of message volume or user count
- Criteria for reliability under network failure or high load
- Specifics of manageability including interface usability and access controls
- Whether audit logs shall be retained for 30 days or longer
- How real-time monitoring thresholds are defined
- Which specific identity-based authentication mechanisms (e.g., MFA, device fingerprinting) should be implemented for email access in the university environment?
- How will dynamic risk-based policies be defined and evaluated to determine whether a user should be allowed to access sensitive data or perform high-risk actions?
- What thresholds will trigger quarantining of messages or generation of security alerts?
- How will real-time monitoring of device health, patch levels, and network traffic be implemented for email endpoints?
- Will the system use certificate-based device identity or other forms of device authentication?
- How will access to sensitive data (e.g., payroll) be controlled based on user behavior and location?
- What is the definition of 'scalability' in terms of message volume or number of users, and how will it be measured during deployment?
- What are the specific reliability criteria under network failure or high load conditions?

- How will manageability be ensured through interface usability, access controls, and operational workflows?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: SEC-001. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

Related requirements: SEC-001. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

No specific interface requirement was supplied; confirmation is required.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall inspect incoming and outgoing email traffic for phishing attempts, malicious attachments, spoofed messages, and suspicious links.

Traceability: E-001, TEST-001, TEST-002

FR-002 · Should · explicit

The system shall allow authorized administrators to review quarantined messages and release them if deemed safe.

Traceability: E-002, TEST-003, TEST-004

3.3 Performance requirements

NFR-001 · Should · explicit

The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.

Traceability: E-004, TEST-005

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

Also covered by: NFR-001.

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: NFR-001.

SEC-001 · Should · explicit

The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Traceability: E-003, TEST-006, TEST-007

3.6.4 Maintainability

Also covered by: NFR-001.

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

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